Spider Dataset Benchmark: Baseline vs. Proposed Algorithm

1. Overview and Objectives

This document records the empirical evaluation of the proposed four-factor, graph-enhanced schema-linking algorithm against the baseline (full schema prompt) across 5 diverse databases from the Spider Text-to-SQL benchmark.

The evaluation focuses on determining whether the algorithm provides the exact required schema to the LLM to generate queries accurately without sending redundant tables and columns. This report presents a direct performance comparison between the **Full-Schema Baseline** and the **Proposed Pipeline** across 5 representative databases from the Spider Text-to-SQL benchmark (hr_1, car_1, flight_2, world_1, and sakila_1), spanning 25 distinct benchmark queries.

The evaluation was executed on SQLite with Gemini (gemini-3.5-flash-lite) under strict 4.6-second request pacing (< 13.0 RPM) to observe true model behavior and execution accuracy without API rate-limit distortion.

2. Selected Databases Profile

The 5 databases were selected to span a wide spectrum of dataset scales, ranging from small tables (100+ rows) up to enterprise-scale tables (16,000+ rows), with structural complexity ranging from 3 to 16 tables:

#	Database	Domain	Table Count	Total Columns	Total Rows	Primary Large Table
1	hr_1	Human Resources	7	35	216	employees (107 Rows)
2	car_1	Automotive Specs	6	23	890	cars_data (406 rows)
3	flight_2	Airlines & Flights	3	13	1,312	flights (1,200 rows)
4	world_1	Global Demographics	3	24	5,302	city (4,079 rows)
5	sakila_1	Media Rental Store	16	89	46,823	payment (16,049 rows)

3. Overall Performance Comparison

Metric	Full-Schema Baseline	Proposed Pipeline	Net Improvement
Prompt Tokens	757.08	361.92	-52.2% context savings (-395.2 tokens)
Columns Sent	36.80	7.84	-73.4% schema clutter reduction
Schema Reduction	0.0%	73.40%	High compression ratio
Algorithm Linking Latency	0.0 ms	36.2 ms (avg)	Fast vectorized in-memory graph linking
LLM Generation Latency	3.00 s	2.90 s	Faster inference due to compact prompt
End-to-End (Total) Latency	3.00 s	2.93 s	Net faster end-to-end than baseline
Schema Linking Precision	9.19%	36.02%	+26.83 points (4x precision increase)
Schema Recall Accuracy	100.0%	92.67%	Near-perfect query answerability
Schema F1-Score (Harmonic Mean of Precision and Recall)	16.37%	46.74%	+30.37 points F1 gain
SQL Execution Success	100.0%	100.0%	100% valid, executable SQLite syntax
Execution Accuracy	68.0% (17/25)	72.0% (18/25)	+4.0% accuracy gain (Proposed beats Baseline)
Generation Throughput (tokens/s ~ tps)	6.84 tps	7.10 tps	Consistent token output throughput

Note: One Time Cold-Start Linking Latency of 26.06s for loading local semantic model into RAM need not be considered.

4. Per-Database Breakdown

Database: hr_1

Metric	Full-Schema Baseline	Proposed Pipeline	Net Improvement
Prompt Tokens	829.20	433.60	-47.7% context savings (-395.6 tokens)
Columns Sent	35.00	10.20	-70.9% schema clutter reduction
Schema Reduction	0.0%	70.90%	High compression ratio
Algorithm Linking Latency	0.0 ms	27.6 ms (warm avg)	Fast vectorized in-memory graph linking (26.06s cold load)
LLM Generation Latency	3.22 s	2.98 s	Faster inference due to compact prompt (-0.24 s)
End-to-End (Total) Latency	3.22 s	3.01 s (warm)	Net faster end-to-end than baseline (-0.21 s)
Schema Linking Precision	9.70%	32.20%	+22.50 points (3.3x precision increase)
Schema Recall Accuracy	100.0%	93.30%	Preserved query answerability (93.3% recall)
Schema F1-Score (Harmonic Mean of Precision and Recall)	17.70%	47.50%	+29.80 points F1 gain
SQL Execution Success	100.0%	100.0%	100% valid, executable SQLite syntax
Execution Accuracy	40.0% (2/5)	60.0% (3/5)	+20.0% accuracy gain (Proposed beats Baseline: 3/5 vs 2/5)
Generation Throughput (tokens/s ~ tps)	8.60 tps	8.36 tps	Consistent token output throughput

Database: car_1

Metric	Full-Schema Baseline	Proposed Pipeline	Net Improvement
Prompt Tokens	512.40	325.80	-36.4% context savings (-186.6 tokens)
Columns Sent	23.00	6.60	-71.3% schema clutter reduction
Schema Reduction	0.0%	71.30%	High compression ratio
Algorithm Linking Latency	0.0 ms	33.5 ms (avg)	Fast vectorized in-memory graph linking
LLM Generation Latency	2.72 s	2.84 s	Comparable inference latency (+0.12 s)
End-to-End (Total) Latency	2.72 s	2.87 s	Comparable end-to-end turnaround (+0.15 s)
Schema Linking Precision	11.30%	69.20%	+57.90 points (6.1x precision increase)

Schema Recall Accuracy	100.0%	100.0%	100% complete schema recall
Schema F1-Score (Harmonic Mean of Precision and Recall)	20.20%	75.00%	+54.80 points F1 gain
SQL Execution Success	100.0%	100.0%	100% valid, executable SQLite syntax
Execution Accuracy	100.0% (5/5)	100.0% (5/5)	100% execution accuracy (All 5 queries passed)
Generation Throughput (tokens/s ~ tps)	9.18 tps	8.60 tps	Consistent token output throughput

Database: flight_2

Metric	Full-Schema Baseline	Proposed Pipeline	Net Improvement
Prompt Tokens	375.20	289.60	-22.8% context savings (-85.6 tokens)
Columns Sent	13.00	5.20	-60.0% schema clutter reduction
Schema Reduction	0.0%	60.00%	High compression ratio
Algorithm Linking Latency	0.0 ms	32.4 ms (avg)	Fast vectorized in-memory graph linking
LLM Generation Latency	2.94 s	2.90 s	Faster inference due to compact prompt (-0.04 s)
End-to-End (Total) Latency	2.94 s	2.93 s	Net faster end-to-end than baseline (-0.01 s)
Schema Linking Precision	16.90%	39.00%	+22.10 points (2.3x precision increase)
Schema Recall Accuracy	100.0%	80.00%	80.0% schema recall (Compact anchor preservation)
Schema F1-Score (Harmonic Mean of Precision and Recall)	28.80%	50.50%	+21.70 points F1 gain
SQL Execution Success	100.0%	100.0%	100% valid, executable SQLite syntax
Execution Accuracy	60.0% (3/5)	60.0% (3/5)	Identical execution accuracy (3/5 queries passed)
Generation Throughput (tokens/s ~ tps)	4.30 tps	4.44 tps	Consistent token output throughput

Database: world_1

Metric	Full-Schema Baseline	Proposed Pipeline	Net Improvement
Prompt Tokens	517.80	303.80	-41.3% context savings (-214.0 tokens)
Columns Sent	24.00	5.20	-78.3% schema clutter reduction
Schema Reduction	0.0%	78.30%	High compression ratio
Algorithm Linking Latency	0.0 ms	33.4 ms (avg)	Fast vectorized in-memory graph linking
LLM Generation Latency	2.91 s	2.77 s	Faster inference due to compact prompt (-0.14 s)
End-to-End (Total) Latency	2.91 s	2.80 s	Net faster end-to-end than baseline (-0.11 s)
Schema Linking Precision	6.70%	29.70%	+23.00 points (4.4x precision increase)
Schema Recall Accuracy	100.0%	90.00%	90.0% schema recall preserved
Schema F1-Score (Harmonic Mean of Precision and Recall)	12.40%	42.70%	+30.30 points F1 gain
SQL Execution Success	100.0%	100.0%	100% valid, executable SQLite syntax
Execution Accuracy	60.0% (3/5)	60.0% (3/5)	Identical execution accuracy (3/5 queries passed)
Generation Throughput (tokens/s ~ tps)	4.70 tps	4.90 tps	Consistent token output throughput

Database: sakila_1

Metric	Full-Schema Baseline	Proposed Pipeline	Net Improvement
Prompt Tokens	1550.80	456.80	-70.5% context savings (-1094.0 tokens)
Columns Sent	89.00	12.00	-86.5% schema clutter reduction
Schema Reduction	0.0%	86.50%	Highest compression ratio on largest schema
Algorithm Linking Latency	0.0 ms	52.2 ms (avg)	Fast vectorized in-memory graph linking
LLM Generation Latency	3.19 s	3.00 s	Faster inference due to compact prompt (-0.19 s)
End-to-End (Total) Latency	3.19 s	3.05 s	Net faster end-to-end than baseline (-0.14 s)
Schema Linking Precision	1.30%	10.00%	+8.70 points (7.7x precision increase)
Schema Recall Accuracy	100.0%	100.00%	100% complete schema recall

Schema F1-Score (Harmonic Mean of Precision and Recall)	2.70%	18.00%	+15.30 points F1 gain
SQL Execution Success	100.0%	100.0%	100% valid, executable SQLite syntax
Execution Accuracy	80.0% (4/5)	80.0% (4/5)	Identical execution accuracy (4/5 queries passed)
Generation Throughput (tokens/s ~ tps)	7.44 tps	9.22 tps	High token generation throughput (+1.78 tps)